

D-Vision-3D-Reconstruct

Memoir3D · Project Insight & Market Research

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The Core Idea

Every party generates hundreds of photos — scattered across everyone's camera rolls. The problem: no single person saw the whole party. The solution: aggregate all photos from all attendees and use AI to reconstruct the event as a 3D navigable space with a timeline.

The key innovation is the **temporal layer**: photos have EXIF timestamps. By grouping photos into 15-minute windows and training a separate 3D Gaussian Splat for each window — all anchored to the same coordinate frame from a single full-event COLMAP reconstruction — the user can scrub through time and watch the venue transform.

Market Research: Competitor Landscape

Product	What It Does	The Gap vs. Memoir3D
Luma AI	Photo/video → Gaussian Splat. Cloud processing.	Single-person capture only. No timeline, no multi-attendee workflow. Pivoted to real estate.
Polycam	LiDAR + photo scan. 4.7★ / 540K iOS downloads.	Requires self-export or Pro session. No crowd-sourcing, no temporal dimension.
Scaniverse (Nianta)	On-device GS processing. Complete workflow.	Single device. No temporal dimension. No multi-contributor workflow.
KIRI Engine	Photogrammetry-first, freemium.	No timeline, no crowd aggregation. Professional tool.
RealityScan (Epic)	High-quality mesh from photos.	Professional tool only. No event/social use case.

Conclusion: The timeline-scrubber over a crowd-sourced party scene is novel and commercially unoccupied. The main competitive risk is Polycam or Luma AI adding multi-user + temporal features — neither has announced this as of mid-2026.

Academic Research Landscape

The research foundation exists — the product implementation does not. This is the gap to fill.

Research / Tool	Relevance	Status
WildGaussians (NeurIPS 2024)	Handles in-the-wild uncontrolled photos with variable lighting.	Research only, no product.
CSS: Crowd-Sourced GS (arXiv:2409.08562)	Addresses pose-free reconstruction from crowd-sourced images.	Research only, no code released.

InstantSplat (NVLabs)	Pose-free GS in seconds. Architecturally ideal.	CUDA only + non-commercial license
Splatt3R	Zero-shot GS from uncalibrated image pairs.	Non-commercial license
OpenSplat	Metal-native GS training. AGPLv3 license.	✓ USE THIS — runs on Apple Silicon
COLMAP	Gold-standard SfM. Metal-accelerated on macOS.	✓ USE THIS — production ready
SAM2 (Meta)	Segment Anything v2. Apache 2.0. MPS support.	✓ USE THIS — for person masking

Apple Silicon Reality Check

Your Mac mini M-series (24 GB unified memory, Metal GPU) is capable but requires careful tool selection. Most research implementations assume NVIDIA CUDA and are blocked on your hardware.

Library	MPS/Metal Support	Verdict
WildGaussians	No (CUDA 11.8 required)	■ Blocked
InstantSplat	No (CUDA 12.1 required)	■ Blocked
gsplat (official)	No (issue #163 open since Apr 2024)	■ Blocked
Splatfacto-W / nerfstudio GS	Broken (issue #3290)	■ Blocked
COLMAP / pycolmap	Yes — native Metal, brew install	■ Full support
OpenSplat	Yes — Metal-native build	■ Use this for GS
SAM2	Yes — MPS with PYTORCH_ENABLE_MPS_FALLBACK=1	■ Supported
Depth Anything v2 Small	Yes — MPS + Core ML option	■ Supported
hloc (SuperPoint+SuperGlue)	Yes — MPS (slower than CUDA)	■ Supported

Performance Benchmarks (M-series estimates)

Stage	100 photos	300 photos	Notes
COLMAP SIFT extraction	~45 sec	~2.5 min	Scales linearly, CPU-bound
COLMAP exhaustive matching	~2 min	~20 min	$O(n^2)$ bottleneck — use vocab tree >200 photos
hloc SuperGlue matching	~8 min	~60+ min	Higher quality, retrieval pairing recommended
COLMAP incremental mapping	~5 min	~25 min	CPU-bound, bundle adjustment included
SAM2 masking (ViT-B+, MPS)	~5 min	~15 min	~3 sec/image on MPS
OpenSplat 30K iters	~17 min	~17 min	Training time $\approx$ constant vs photo count
Total pipeline	~35 min	~80 min	Per time window, sequential

24 GB unified memory is a meaningful advantage over CUDA GPUs with separate VRAM. OpenSplat uses ~13–19 GB for a 300-photo indoor scene, which fits comfortably. The hard limit is ~2M Gaussians — beyond that, OOM risk is high.

Product Design Highlights

Core User Journeys

- **The Host:** Creates event → shares QR code → collects photos from guests → triggers reconstruction → reviews result → shares final link. System warns about sparse coverage before triggering GPU compute.
- **The Viewer:** Receives link → 8-second cinematic intro flythrough → free exploration → scrubs timeline → sees party evolve in 3D. No account required to view.
- **The Contributor:** Scans QR at venue (or taps link in group chat) → mobile browser, no app install → select photos from camera roll → chunked upload → 'Your 27 photos are in the mix!'

3D Viewer UX Design Principles

Core principle: feel like **Google Maps 3D**, not Blender. Non-technical users already know pinch-to-zoom and drag-to-pan.

- Mouse: left-drag = orbit · scroll = zoom · right-drag = pan · double-click = recenter
- Touch: one-finger drag = orbit · pinch = zoom · two-finger drag = pan
- Timeline: full-width bar, 30-min buckets, density histogram, 0.4s cross-fade between windows
- Play mode: auto-advance one window per 4 seconds — time-lapse of the entire party

Progressive Loading (never show a blank screen)

- 0–1s: Fetch scene.json → render timeline with loading skeletons
- 1–5s: Stream foundation_preview.ply (10% density, <5 MB) → user can already orbit
- 5–20s: Stream full foundation.ply in 2 MB chunks → surfaces solidify
- On demand: time windows loaded on scrubber interaction, ±1 window prefetched

Monetization Strategy

Per-Event Pricing (Consumer)

Tier	Price	Limits	Key Features
Free	\$0	1 event, 50 photos	Memoir3D watermark on exports
Starter	\$9 one-time	500 photos	No watermark, 720p flythrough export
Pro	\$29 one-time	5 events, 2K photos/event	Face blur, 1080p export, password protection
Wedding	\$79 one-time	Unlimited photos	4K export, embed code, priority processing

B2B Subscription — \$99/month

Target: wedding photographers, event management companies, corporate comms teams. Includes: unlimited events, white-label viewer, client delivery portal, API access, 5 team seats. High LTV, low churn — repeat buyers making 20–50 events/year.

À La Carte Add-Ons

- Face blur (\$5) · AI highlight reel 60s (\$8) · Extended storage 5yr (\$12/yr) · Rush processing (\$15) · Vanity URL (\$8)

Realistic Timeline Summary

Scenario	MVP (M6)	Production (M7)	Notes
Optimistic	Week 26 (~6.5 mo)	Week 34 (~8.5 mo)	Everything works first try
Realistic ← use this	Week 32 (~8 mo)	Week 40 (~10 mo)	Normal debugging, a few restarts
Pessimistic	Week 42 (~10.5 mo)	Week 52 (~13 mo)	Timeline feature requires research pivot

The difference between optimistic and pessimistic is almost entirely driven by two things: (1) how long the timeline/photo-Gaussian association problem takes to solve, and (2) whether COLMAP performance on Apple Silicon is acceptable or requires architectural workarounds. Both can be validated in Phase 0 (Weeks 1–3).

The One Rule

Before writing a single line of app code, take 100 photos of your room and run the full COLMAP → OpenSplat pipeline on your Mac mini. The numbers you get — training time, quality, memory usage — will inform every product decision for the next 10 months. No planning document replaces this experiment.

The most important test: does the .ply file render as a recognizable room in Supersplat? If yes, the foundation is proven and you can build with confidence. If no, you have discovered the core blocker before investing months of work.